

CSA0609 - DESIGN AND ANALYSIS OF ALGORITHMS

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DAA LAB EXPERIMENT 1

First Palindromic String in an Array

Aim

To write a Python program to find the **first palindromic string** in an array and analyze its time complexity.

Algorithm

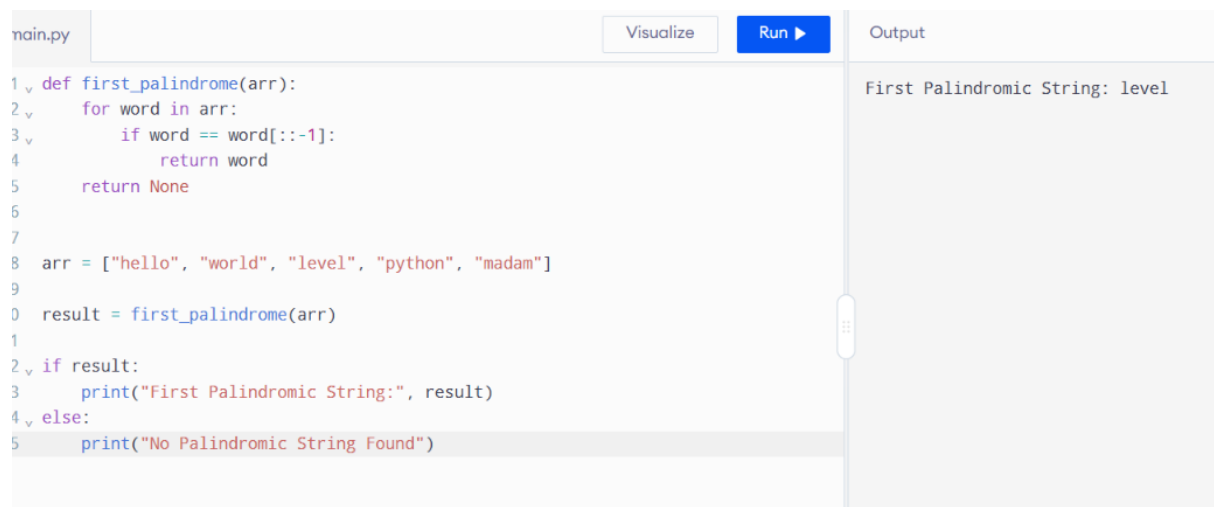
1. Start.
2. Read the array of strings.
3. Traverse each string from left to right.
4. For every string, check whether it is equal to its reverse.
5. If the string is a palindrome, display it and stop searching.
6. If no palindrome is found, display an appropriate message.
7. Stop.

Python Source Code

```
def first_palindrome(arr):
    for word in arr:
        if word == word[::-1]:
            return word
    return None

arr = ["hello", "world", "level", "python", "madam"]
result = first_palindrome(arr)
if result:
    print("First Palindromic String:", result)
else:
    print("No Palindromic String Found")
```

Output



```
main.py Visualize Run ▶ Output
1 def first_palindrome(arr):
2     for word in arr:
3         if word == word[::-1]:
4             return word
5     return None
6
7
8 arr = ["hello", "world", "level", "python", "madam"]
9
0 result = first_palindrome(arr)
1
2 if result:
3     print("First Palindromic String:", result)
4 else:
5     print("No Palindromic String Found")
```

Result

Thus, the Python program to find the first palindromic string in an array was successfully implemented and its complexity was analyzed.